

TECHNICAL MEMORANDUM: IES STANDARD FORMAT FOR THE ELECTRONIC TRANSFER OF SPECTRAL DATA

AN AMERICAN NATIONAL STANDARD

ANSI/IES TM-27-20 +ERRATA 1

TECHNICAL MEMORANDUM: IES STANDARD FORMAT FOR THE ELECTRONIC TRANSFER OF SPECTRAL DATA AN AMERICAN NATIONAL STANDARD

Publication of this report
has been approved by IES.
Suggestions for revisions
should be directed to IES.

**Prepared for IES by
The IES Computer Committee**



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1.0 Introduction and Scope

1.1 Introduction

The architectural lighting community has long relied on standardized file formats for the electronic transfer of photometric data and related information. These file formats include ANSI/IES LM-63-19,¹ CIBSE TM-14,² and EULUMDAT.³

Recognizing the need to distribute non-photometric data related to luminaires, the IES Computer Committee developed *IES LM-74-05/R2009, IES Standard File Format for the Electronic Transfer of Luminaire Component Data*.⁴

There is, however, also a need to distribute spectral data. Such data are pertinent, but not limited to, the spectral radiant flux of lamps and other light sources, the spectral transmittance of color filters, and the spectral reflectance of paint and other finishes. These data may be obtained when testing a light source, luminaire, or material and may then be used in lighting application software, rendering software, analysis software, or optical design software.

Generally, the document will contain a single spectral power distribution. Future formats that contain multiple spectra, including spectra as a function of angle, are planned, with this document being a stepping stone in that direction, while also creating a format that still has widespread usability across industries.

The data are formatted in accordance with W3C Extensible Markup Language (XML) 1.1 Recommendation and the W3C XML Schema Definition Language (XSD) (see **Section 2**). This allows end users to utilize and view the data directly without the need for proprietary software.

In the event of any ambiguity or discrepancy with respect to the description of XML or XML Schema in this document, the W3C Recommendations take precedence.

1.2 Scope

This document specifies an electronic (XML based) data format for the transfer of spectral data. This document may be used for the transfer of spectral data of optical radiation including light sources, lamps, and luminaires, as well as reflectance and transmittance spectra of materials. The document is limited to containing a single spectral data set.

Details about the XML document format, XML Schema, XSLT transforms and more can be found at the W3C's website, the authority for the XML document format. This document is intended as a description of a specific implementation of an XML document and not as a tutorial on the XML document format itself.

2.0 Normative References

2.1 ANSI/IES LS-1-20

Illuminating Engineering Society Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020.

2.2 CIE 15:2004

International Commission on Illumination (CIE). Colorimetry. Vienna: CIE; 2004.

2.3 XML

Extensible Markup Language (XML) 1.0, 5th ed. W3C. Online: <http://www.w3.org/TR/REC-xml>. (Accessed 2019 Jul 19).

2.4 XML XSD Part 1

Schema Definition Language (XSD) 1.1, Part 1: Structures. W3C. Online: <http://www.w3.org/TR/xmlschema11-1>. (Accessed 2019 Jul 19).

2.5 XML XSD Part 2

Schema Definition Language (XSD) 1.1, Part 2: Datatypes. W3C. Online: <http://www.w3.org/TR/xmlschema11-2>. (Accessed 2019 Jul 19).

3.0 Definitions

3.1 bandpass correction

The correction of measured spectral data to account for the bandpass width of the detector element and the scanning interval. There are several methods of bandpass correction, and this correction is usually performed either in the measurement equipment itself or by the data processing software.

3.2 document

Instead of using *file* as the description for this data storage mechanism, the term *document* will be used. *Document* is more appropriate when the information may not actually exist as a physical file, but may be generated dynamically and transmitted.

3.3 spectral data

Radiometric power as a function of wavelength, typically represented as a table of wavelength and corresponding radiometric power.

4.0 XML Schema for Spectral Data Transfer

Descriptions of the intended use of each XML element and attribute in the schema are contained in **Sections 4.1** and **4.2**.

The spectral power distribution XML document is defined in terms of the W3C XML Schema presented in **Section 4.3**.

The XML document is separated into a Header element (see **Section 4.1**) and a Spectral Distribution element (see **Section 4.2**). The Header element contains all generic information about the dataset; the Spectral Distribution element contains the specific data.

4.1 Header Element

The Header element is the parent of the document header section.

This section contains information that is not specific to the spectral data. Many of the header elements are optional. The intent is for the document to contain as much header information as possible so that the origin and purpose of the document is clear in the future. Specific elements may not be appropriate for a given document. For example, a document containing stellar spectra would not need the Manufacturer element, but may still have the Laboratory element. Header elements are listed in **Table 4-1**.

Table 4-1. Header Element Fields

Element Description	Element Name	Data Type	Required	Document Section
Manufacturer	Manufacturer	String	No	4.1.1
Catalog Number	CatalogNumber	String	No	4.1.2
Description	Description	String	Yes	4.1.3
Document Creator	DocumentCreator	String	Yes	4.1.4
Unique Identifier	UniquelIdentifier	String	No	4.1.5
Measurement Equipment	MeasurementEquipment	String	No	4.1.6
Laboratory	Laboratory	String	No	4.1.7
Report Number	ReportNumber	String	No	4.1.8
Report Date	ReportDate	XML DateTime	No	4.1.9
Document Creation Date	DocumentCreationDate	XML DateTime	Yes	4.1.10
Comments	Comments	String	No	4.1.11

4.1.1 Manufacturer Element. The optional Manufacturer element identifies the manufacturer of the device under test.

4.1.2 Catalog Number Element. The optional CatalogNumber element identifies the manufacturer's product catalog number.

4.1.3 Description Element. The required Description element contains a text description of the spectral data in the document.

4.1.4 Document Creator Element. The required DocumentCreator element identifies the creator of the document, which may be a test lab, a research group, a standard body, a company, or an individual.

4.1.5 Unique Identifier Element. The optional UniquelIdentifier element contains a unique identifier for the product under test or the spectral data in the document.

4.1.6 Measurement Equipment Element. The optional MeasurementEquipment element contains a description of the equipment used to measure the spectral data.

4.1.7 Laboratory Element. The optional Laboratory element identifies the testing laboratory name that performed the spectral data measurements. If the data is generated and not tested at a laboratory, this field shall contain the name of the company generating the data.

4.1.8 Report Number Element. The optional ReportNumber element identifies the testing laboratory report number.

4.1.9 Report Date Element. The optional ReportDate element identifies the testing laboratory report date using the XML DateTime Data Type: YYYY-MM-DDThh:mm:ss.

4.1.10 Document Creation Date Element. The required DocumentCreationDate element identifies the document creation date using the XML DateTime Data Type: YYYY-MM-DDThh:mm:ss.

4.1.11 Comments Element. The optional Comments element provides additional information relating to the tested and reported data.

4.2 Spectral Distribution Element

The SpectralDistribution element is the parent of the spectral distribution data. This element contains information that is specific to the spectral data. Elements are detailed in **Table 4-2**.

Table 4-2. Spectral Distribution Fields

Element Description	Element Name	Data Type	Required	Document Section
Spectral Quantity	SpectralQuantity	String	Yes	4.2.1
Reflection Geometry	ReflectionGeometry	String	Yes, only if Spectral Quantity = "reflectance"	4.2.2
Transmission Geometry	TransmissionGeometry	String	Yes, only if Spectral Quantity = "transmittance"	4.2.3
Bandwidth FWHM	BandwidthFWHM	Decimal	No	4.2.4
Bandwidth Corrected	BandwidthCorrected	Boolean	No	4.2.5
Spectral Data	SpectralData	String	Yes	4.2.6

4.2.1 Spectral Quantity Element. The SpectralQuantity attribute provides the quantity of measurement for each element of the spectral data. Valid SpectralQuantity attributes are identified in **Table 4-3**.

Attribute	Spectral Quantity	Symbols	Units
"flux"	Radiant flux	$\Phi_{\lambda} = \frac{d\Phi}{d\lambda}$	W·nm ⁻¹
"absorptance"	Absorptance	$\alpha(\lambda) = \frac{\Phi_{a,\lambda}}{\Phi_{i,\lambda}}$	
"transmittance"	Transmittance	$\tau(\lambda) = \frac{\Phi_{t,\lambda}}{\Phi_{i,\lambda}}$	
"reflectance"	Reflectance	$\rho(\lambda) = \frac{\Phi_{r,\lambda}}{\Phi_{i,\lambda}}$	
"intensity"	Radiant intensity	$I_{\lambda} = \frac{dI}{d\lambda}$	W·sr ⁻¹ ·nm ⁻¹
"irradiance"	Irradiance	$E_{\lambda} = \frac{dE}{d\lambda}$	W·m ⁻² ·nm ⁻¹
"radiance"	Radiance	$L_{\lambda} = \frac{d^2\Phi}{d\omega(dA \times \cos\theta)d\lambda}$	W·m ⁻² ·nm ⁻¹ ·sr ⁻¹
"exitance"	Radiant exitance	$M_{\lambda} = \frac{dM}{d\lambda}$	W·m ⁻² ·nm ⁻¹
"R-Factor"	Reflectance factor	$R_{\lambda} = \frac{dR}{d\lambda}$	nm ⁻¹
"T-Factor"	Transmittance factor	$T_{\lambda} = \frac{dT}{d\lambda}$	nm ⁻¹
"relative"	Relative spectral power distribution		nm ⁻¹
"other"	User-specified in Comments in 4.1.11		User-specified in Comments in 4.1.11

All wavelengths shall be specified in nanometers. Relative measurements (i.e., absorptance, transmittance, reflectance, reflectance factor, transmittance factor, and relative spectral power distribution) shall be specified in the range 0 to 1.0, not as a percentage.

4.2.2 Reflection Geometry Element. Spectral reflectance factors require specification of the geometric conditions. Valid ReflectionGeometry attributes in accordance with CIE 15:2004, Colorimetry (refer to **Section 2.2**) are in **Table 4-4**.

Attribute	Geometry
"di:8"	Diffuse / eight-degree, specular component included
"de:8"	Diffuse / eight-degree, specular component excluded
"8:di"	Eight-degree / diffuse, specular component included
"8:de"	Eight-degree / diffuse, specular component excluded
"d:d"	Diffuse / diffuse
"d:0"	Alternative diffuse
"45a:0"	Forty-five degree annular / normal
"45c:0"	Forty-five degree circumferential / normal
"0:45a"	Normal / forty-five degree annular
"45x:0"	Forty-five degree directional / normal
"0:45x"	Normal / forty-five degree directional
"other"	User-specified in Comments in 4.1.11

4.2.3 Transmission Geometry Element. Spectral transmittance factors require specification of the geometric conditions. Valid TransmissionGeometry attributes in accordance with CIE 15:2004, Colorimetry (See **Section 2.2**) are in **Table 4-5**.

Attribute	Geometry
"0:0"	Normal / normal
"di:0"	Diffuse / normal, regular component included
"de:0"	Diffuse / normal, regular component excluded
"0:di"	Normal / diffuse, regular component included
"0:de"	Normal / diffuse, regular component excluded
"d:d"	Diffuse / diffuse
"other"	User-specified in Comments in 4.1.11

4.2.4 Bandwidth FWHM Element. The optional BandwidthFWHM attribute specifies the spectroradiometer full-width half-maximum bandwidth in nanometers.

4.2.5 Bandwidth Corrected Element. The optional BandwidthCorrected attribute specifies that bandwidth correction has been applied to the measured data.

If necessary, the bandwidth correction method should be documented in the optional Comments element (**Section 4.1.11**).

4.2.6 Spectral Data. The SpectralData element specifies the spectral data for the identified wavelength. The element contains the value to be reported, and there is an attribute for the wavelength at which this value occurs.

4.3 XML Schema

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<iesxsd:schema xmlns:iesxsd="http://www.ies.org/iestm2714Schema">
  <iesxsd:element name="iestm2714">
    <iesxsd:complexType>
      <iesxsd:all>
        <iesxsd:element name="Header">
          <iesxsd:complexType>
            <iesxsd:all>
              <iesxsd:element name="Manufacturer" type="iesxsd:string" minOccurs="0"/>
              <iesxsd:element name="CatalogNumber" type="iesxsd:string" minOccurs="0"/>
              <iesxsd:element name="Description" type="iesxsd:string"/>
              <iesxsd:element name="DocumentCreator" type="iesxsd:string"/>
              <iesxsd:element name="UniqueIdentifier" type="iesxsd:string" minOccurs="0"/>
              <iesxsd:element name="MeasurementEquipment" type="iesxsd:string" minOccurs="0"/>
              <iesxsd:element name="Laboratory" type="iesxsd:string" minOccurs="0"/>
              <iesxsd:element name="ReportNumber" type="iesxsd:string" minOccurs="0"/>
              <iesxsd:element name="ReportDate" type="iesxsd:date" minOccurs="0"/>
              <iesxsd:element name="DocumentCreationDate" type="iesxsd:date"/>
              <iesxsd:element name="Comments" type="iesxsd:string" minOccurs="0"/>
            </iesxsd:all>
          </iesxsd:complexType>
        </iesxsd:element>
        <iesxsd:element name="SpectralDistribution">
          <iesxsd:complexType>
            <iesxsd:all>
              <iesxsd:element name="SpectralQuantity" minOccurs="1">
                <iesxsd:simpleType>
                  <iesxsd:restriction base="iesxsd:string" use="required">
                    <iesxsd:enumeration value="flux"/>
                    <iesxsd:enumeration value="absorptance"/>
                    <iesxsd:enumeration value="transmittance"/>
                    <iesxsd:enumeration value="reflectance"/>
                    <iesxsd:enumeration value="intensity"/>
                    <iesxsd:enumeration value="irradiance"/>
                    <iesxsd:enumeration value="radiance"/>
                    <iesxsd:enumeration value="exitance"/>
                    <iesxsd:enumeration value="R-factor"/>
                    <iesxsd:enumeration value="T-factor"/>
                    <iesxsd:enumeration value="relative"/>
                    <iesxsd:enumeration value="other"/>
                  </iesxsd:restriction>
                </iesxsd:simpleType>
              </iesxsd:element>
              <iesxsd:element name="ReflectionGeometry" minOccurs="0">
                <iesxsd:simpleType>
                  <iesxsd:restriction base="iesxsd:string" use="required">
```

```

        <iesxsd:enumeration value="di:8"/>
        <iesxsd:enumeration value="de:8"/>
        <iesxsd:enumeration value="8:di"/>
        <iesxsd:enumeration value="8:de"/>
        <iesxsd:enumeration value="d:d"/>
        <iesxsd:enumeration value="d:0"/>
        <iesxsd:enumeration value="45a:0"/>
        <iesxsd:enumeration value="45c:0"/>
        <iesxsd:enumeration value="0:45a"/>
        <iesxsd:enumeration value="45x:0"/>
        <iesxsd:enumeration value="0:45x"/>
        <iesxsd:enumeration value="other"/>
    </iesxsd:restriction>
</iesxsd:simpleType>
</iesxsd:element>
<iesxsd:element name="TransmissionGeometry" minOccurs="0">
    <iesxsd:simpleType>
        <iesxsd:restriction base="iesxsd:string" use="required">
            <iesxsd:enumeration value="0:0"/>
            <iesxsd:enumeration value="di:0"/>
            <iesxsd:enumeration value="de:0"/>
            <iesxsd:enumeration value="0:di"/>
            <iesxsd:enumeration value="0:de"/>
            <iesxsd:enumeration value="d:d"/>
            <iesxsd:enumeration value="other"/>
        </iesxsd:restriction>
    </iesxsd:simpleType>
</iesxsd:element>
<iesxsd:element name="BandwidthFWHM" type="iesxsd:decimal" minOccurs="0"/>
<iesxsd:element name="BandwidthCorrected" type="iesxsd:boolean" minOccurs="0"/>
<iesxsd:element name="SpectralData" type="iesxsd:decimal" minOccurs="1" maxOccurs="unbounded">
    <iesxsd:attribute name="wavelength" type="iesxsd:decimal" use="required"/>
</iesxsd:element>
</iesxsd:all>
</iesxsd:complexType>
</iesxsd:element>
</iesxsd:all>
</iesxsd:complexType>
</iesxsd:element>
</iesxsd:schema>

```

5.0 Branding – Logo and Name

The major goal of this document is to create a unified, industry standard format for spectral data transfer. Given that there are many existing files in the industry, each used only for a specific piece of software, confusion about the new unified format may arise.

To help eliminate this confusion, spectral data transfer conforming to this document shall always be referred to as either an “IES TM-27 SPDY document” or simply an “SPDY document.” Whenever the document is displayed for download or distribution it shall be shown with “IES-TM-27” as a descriptor, and the document extension of “.spdy” shall be used.

A logo and icon can be used to identify an IES-TM-27 SPDY document. This can be applied as an icon for the .spdy document type by any software that uses SPDY documents; it can also be used on websites and in publications to show either that the documents are IES-TM-27 SPDY documents or that the software supports the TM-27 SPDY document format. The logo is shown in **Figure 5-1** and the icon in **Figure 5-2**. Both are released under the CC0 “No Rights Reserved” licensing scheme into the public domain. Vector logo files can be obtained from the Marketing Department at the IES offices in New York.



Figure 5-1. TM-27 Logo



Figure 5-2. TM-27 Icon

GENERAL REFERENCES

- 1 Illuminating Engineering Society. *ANSI/IESNA LM-63-19, IES Standard File Format for the Electronic Transfer of Photometric Data and Related Information*. New York: IES; 2019.
- 2 *EULUMDAT File Format Specification*. (Note: Original file format developed by Dr. Axel Stockmar, Light Consult Inc., Berlin, 1990). Online: www.helios32.com/Eulumdat.htm. (Accessed 2019 Jul 19).
- 3 Chartered Institution of Building Services Engineers. *TM14, Technical Memorandum: Standard File Format for the Electronic Transfer of Luminaire Photometric Data*. London: CIBSE; 1988.
- 4 Illuminating Engineering Society. *IES LM-74-05, IESNA Standard File Format for the Electronic Transfer of Luminaire Component Data*. New York: IES; 2005.

Annex A – Sample XML Document

The following XML is a compliant example of a spectral data document.

```
<?xml version="1.0"?>
<?xml-stylesheet type="text/xsl" href="TM-27-14.xsl"?>
<IESTM2714 xmlns="http://www.ies.org/iestm2714" version="1.0">
  <Header>
    <Manufacturer>Academy Lighting</Manufacturer>
    <CatalogNumber>XET 55529</CatalogNumber>
    <Description>LED 2' x 4' Troffer</Description>
    <DocumentCreator>Apex Analytics</DocumentCreator>
    <Laboratory>Apex Analytics</Laboratory>
    <UniqueIdentifier>TS3005k3d96</UniqueIdentifier>
    <ReportNumber>APEX-091101-004</ReportNumber>
    <ReportDate>2009-11-01</ReportDate>
    <DocumentCreationDate>2011-11-21</DocumentCreationDate>
    <Comments>Ambient temperature 25 degrees C.</Comments>
  </Header>
  <SpectralDistribution>
    <SpectralQuantity>relative</SpectralQuantity>
    <BandwidthFWHM>2.0</BandwidthFWHM>
    <BandwidthCorrected>true</BandwidthCorrected>
    <SpectralData wavelength="400.0">0.47</SpectralData>
    <SpectralData wavelength="410.0">0.51</SpectralData>
    <SpectralData wavelength="420.0">0.56</SpectralData>
    <SpectralData wavelength="430.0">0.61</SpectralData>
    <SpectralData wavelength="440.0">0.66</SpectralData>
    <SpectralData wavelength="450.0">0.72</SpectralData>
    <SpectralData wavelength="460.0">0.76</SpectralData>
    <SpectralData wavelength="470.0">0.77</SpectralData>
    <SpectralData wavelength="480.0">0.79</SpectralData>
    <SpectralData wavelength="490.0">0.81</SpectralData>
    <SpectralData wavelength="500.0">0.83</SpectralData>
    <SpectralData wavelength="510.0">0.85</SpectralData>
    <SpectralData wavelength="520.0">0.88</SpectralData>
    <SpectralData wavelength="530.0">0.92</SpectralData>
    <SpectralData wavelength="540.0">0.05</SpectralData>
    <SpectralData wavelength="550.0">0.99</SpectralData>
    <SpectralData wavelength="560.0">0.99</SpectralData>
    <SpectralData wavelength="570.0">1.00</SpectralData>
    <SpectralData wavelength="580.0">0.95</SpectralData>
    <SpectralData wavelength="590.0">0.91</SpectralData>
    <SpectralData wavelength="600.0">0.76</SpectralData>
    <SpectralData wavelength="610.0">0.62</SpectralData>
    <SpectralData wavelength="620.0">0.51</SpectralData>
    <SpectralData wavelength="630.0">0.40</SpectralData>
    <SpectralData wavelength="640.0">0.29</SpectralData>
    <SpectralData wavelength="650.0">0.21</SpectralData>
  </SpectralDistribution>
</IESTM2714>
```

Annex B – Sample XSL For CSV Presentation

The following XSL illustrates an XML transform that can be used to display the data from a spectral data document in a CSV format.

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<xsl:stylesheet version="1.0"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform"
xmlns:ies="http://www.ies.org/iestm2714">
  <xsl:output method="text" omit-xml-declaration="yes" indent="no"/>
  <xsl:template match="/">
    <xsl:for-each select="ies:IESTM2714/ies:SpectralDistribution/ies:SpectralData">
      <xsl:value-of select="@wavelength"/>
      <xsl:text>,</xsl:text>
      <xsl:value-of select="."/>
      <xsl:text> &#13;</xsl:text>
    </xsl:for-each>
  </xsl:template>
</xsl:stylesheet>
```

Annex C – Sample XSL for HTML Presentation

The following XSL illustrates an XML transform that can be used to display the data from a spectral data document in HTML format.

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<xsl:stylesheet version="1.0"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform"
xmlns:ies="http://www.ies.org/iestm2714">
<xsl:template match="/">
  <html>
  <body>
    <h2>Report Information</h2>
    <table border="1">
      <tr bgcolor="#9acd32">
        <th>Field</th>
        <th>Data</th>
      </tr>
      <xsl:for-each select="ies:IESTM2714/ies:Header/node()">
        <tr>
          <td bgcolor="#9acd32"><xsl:value-of select="name()"/></td>
          <td><xsl:value-of select="."/></td>
        </tr>
      </xsl:for-each>
    </table>
    <h2>Data Table</h2>
    <table border="1">
      <tr bgcolor="#9acd32">
        <th>Wavelength</th>
        <th>Spectral Data</th>
      </tr>
      <xsl:for-each select="ies:IESTM2714/ies:SpectralDistribution/ies:SpectralData">
        <tr>
          <td><xsl:value-of select="@wavelength"/></td>
          <td><xsl:value-of select="."/></td>
        </tr>
      </xsl:for-each>
    </table>
  </body>
</html>
</xsl:template>
</xsl:stylesheet>
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